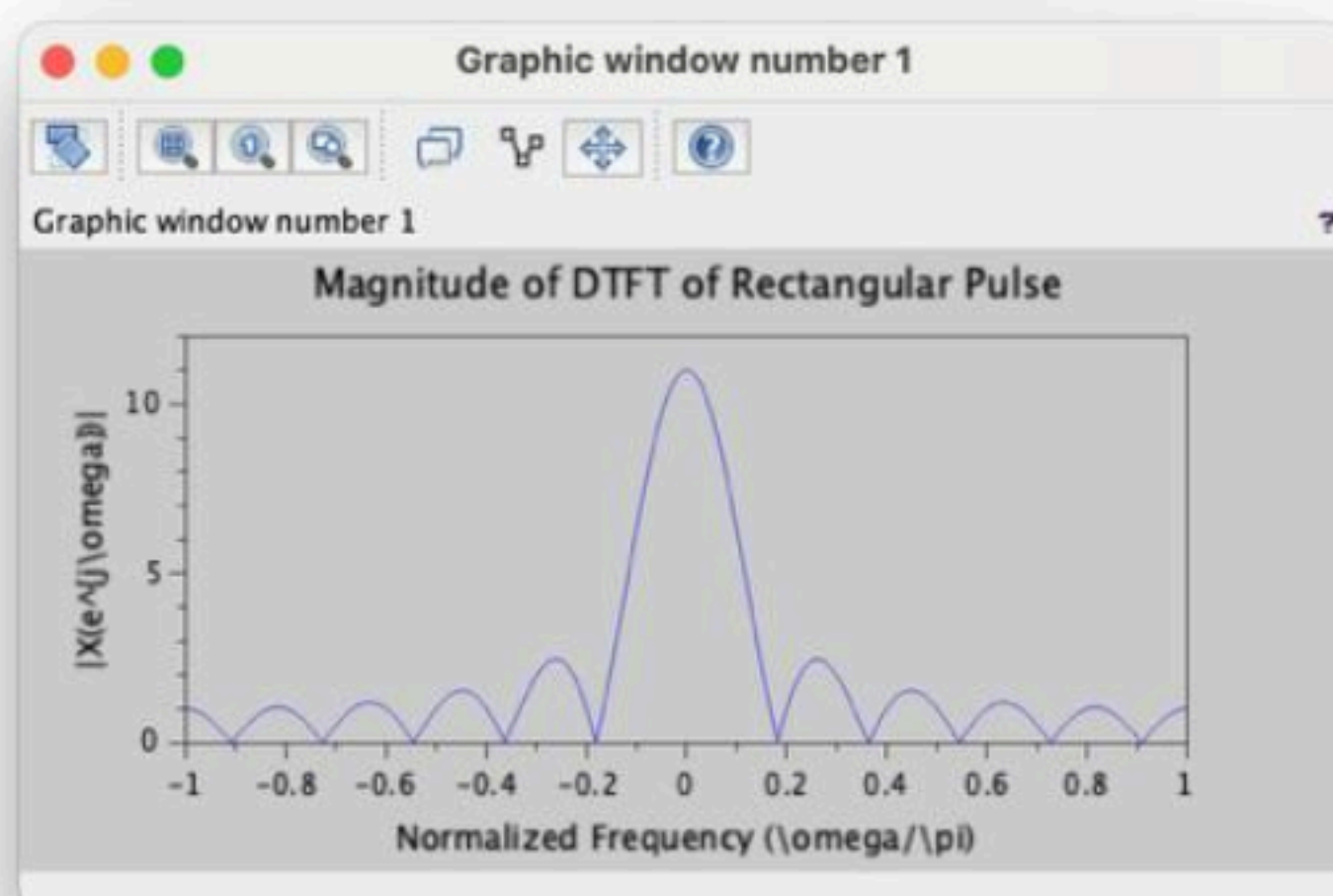
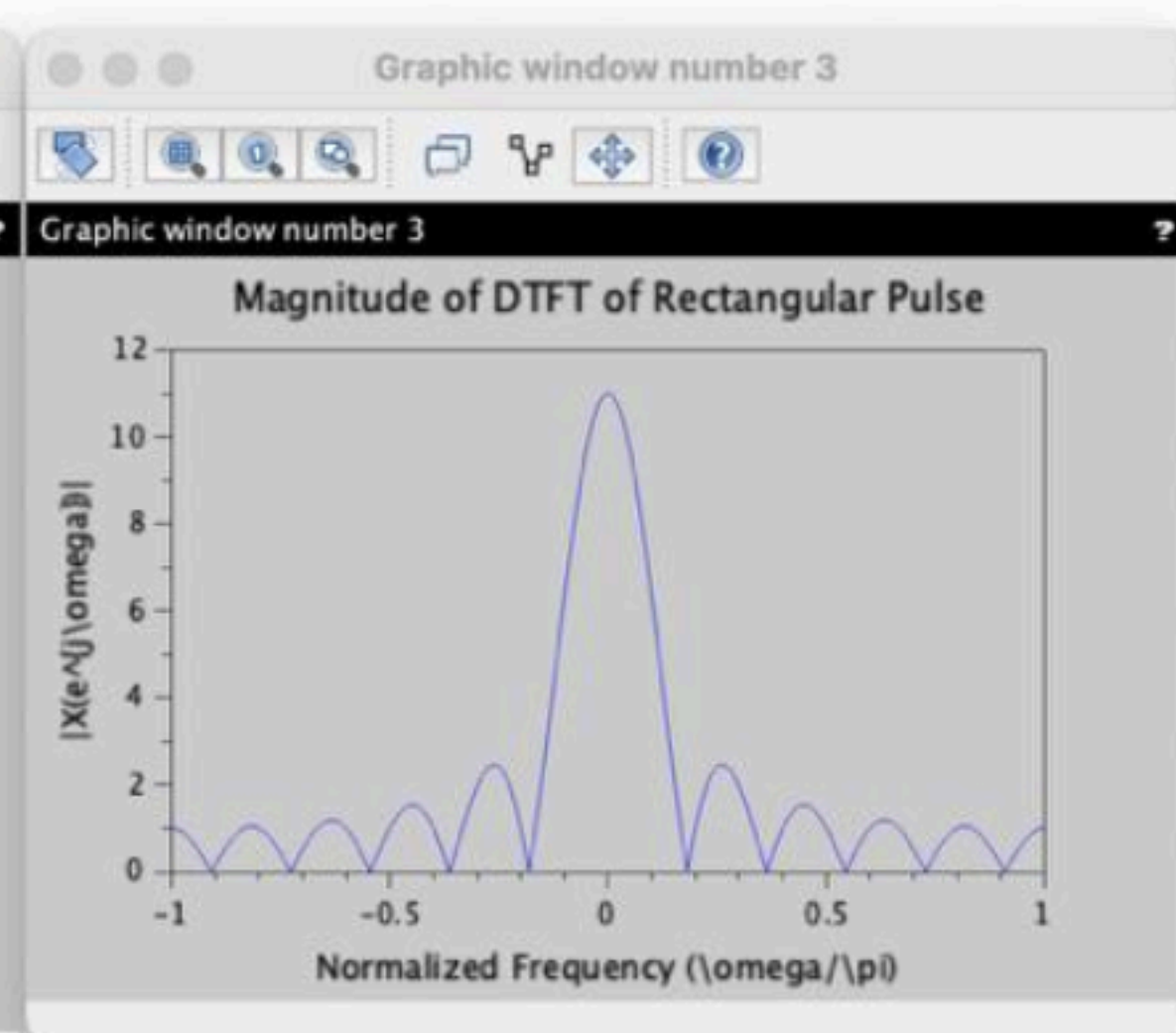
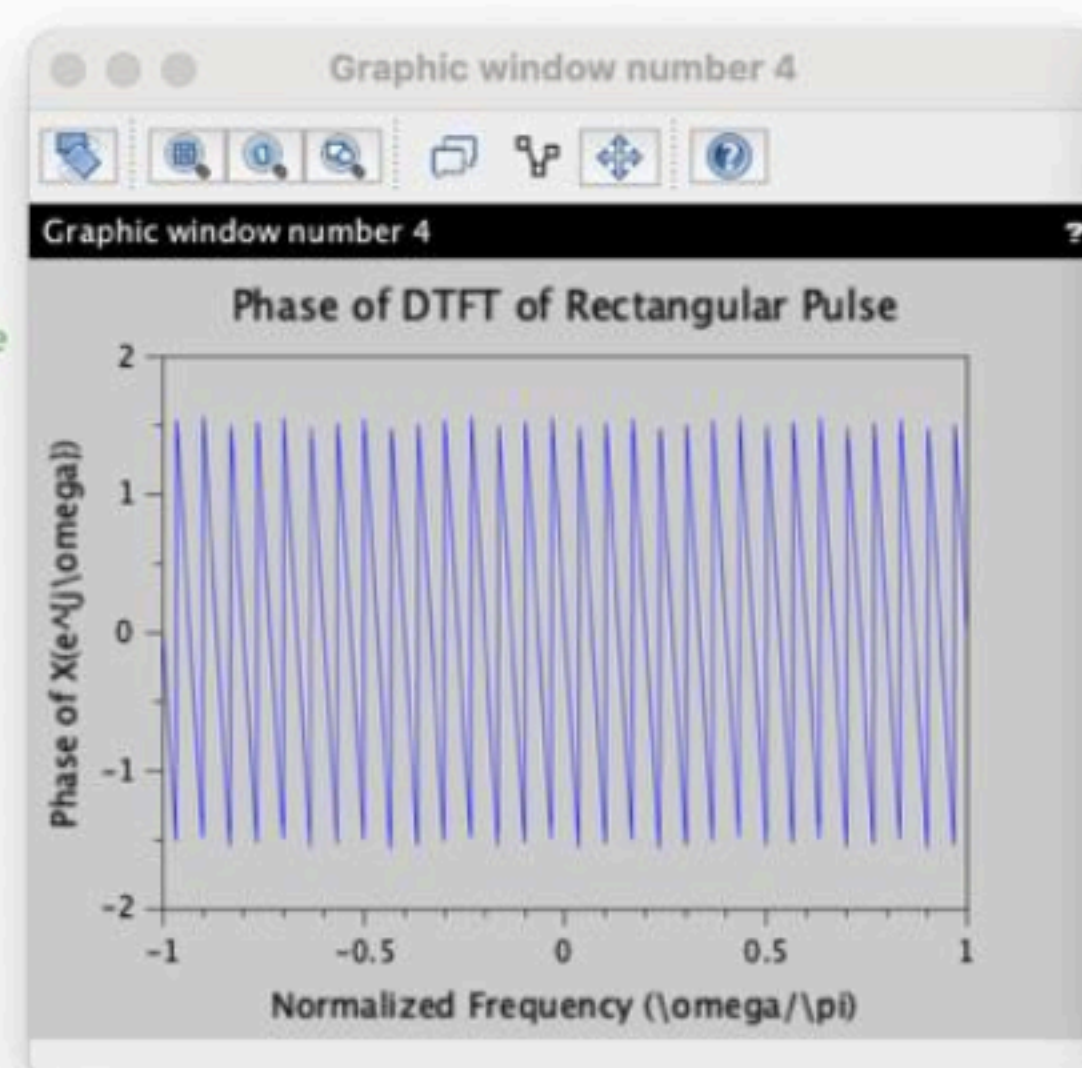


signals 5.2.sce x signals 6.1.sce x 6.2.sce x signals 7.1.sci x 7.2.sce x 8.1.sce x 8.2.sci x 9.1.sce x Untitled 1 x signals 1.2.sce x signals 3.4.sce x

```

1 // Define parameters
2 N = 31; n = 0:N-1; n1 = 10; n2 = 20; // Total number of samples
3 // Sample index
4 // Start of rectangular pulse
5 // End of rectangular pulse
6 // Define the rectangular pulse x[n]
7 x = zeros(1, N); x(n1+1:n2+1) = 1; // Initialize the signal with zeros
8 // Set values between n1 and n2 to 1
9 // Compute DTFT
10 omega = linspace(-%pi, %pi, 1000); X = x * exp(-%i * n' * omega); // Frequency range
11 // DTFT calculation
12 // Plot the magnitude of DTFT
13 figure();
14 plot(omega/%pi, abs(X));
15 xlabel('Normalized Frequency (\omega/\pi)');
16 ylabel('|X(e^{j\omega})|');
17 title('Magnitude of DTFT of Rectangular Pulse');
18 // Plot the phase of DTFT
19 figure();
20 plot(omega/%pi, atan(imag(X)./real(X)));
21 xlabel('Normalized Frequency (\omega/\pi)');
22 ylabel('Phase of X(e^{j\omega})');
23 title('Phase of DTFT of Rectangular Pulse');
24 // Analyze the results in the Scilab console or report
25

```



signals4.1.sce X signals 4.2.sce X signals 5.1.sce X signals 5.2.sce X signals 6.1.sce X 6.2.sce X signals 7.1.sci X 7.2.sce X 8.1.sce X 8.2.sci X

```
1 // Define parameters
2 N = 31; n = 0:N-1; n1 = 10; n2 = 20; // Total number of samples
3 // Sample index
4 // Start of rectangular pulse
5 // End of rectangular pulse
6 // Define the rectangular pulse x[n]
7 x = zeros(1, N); x(n1+1:n2+1) = 1; // Initialize the signal with zeros
8 // Set values between n1 and n2 to 1
9 // Compute DTFT
10 omega = linspace(-pi, pi, 1000); X = x * exp(-i * n * omega); // Frequency range
11 // DTFT calculation
12 // Plot the magnitude of DTFT
13 figure();
14 plot(omega/pi, abs(X));
15 xlabel('Normalized Frequency (\omega/\pi)');
16 ylabel('|X(e^{j\omega})|');
17 title('Magnitude of DTFT of Rectangular Pulse');
18 // Plot the phase of DTFT
19 figure();
20 plot(omega/pi, atan(imag(X)./real(X)));
21 xlabel('Normalized Frequency (\omega/\pi)');
22 ylabel('Phase of X(e^{j\omega})');
23 title('Phase of DTFT of Rectangular Pulse');
24
```

